



Lesson Aim

Explain how to make fortified wines and spirits.

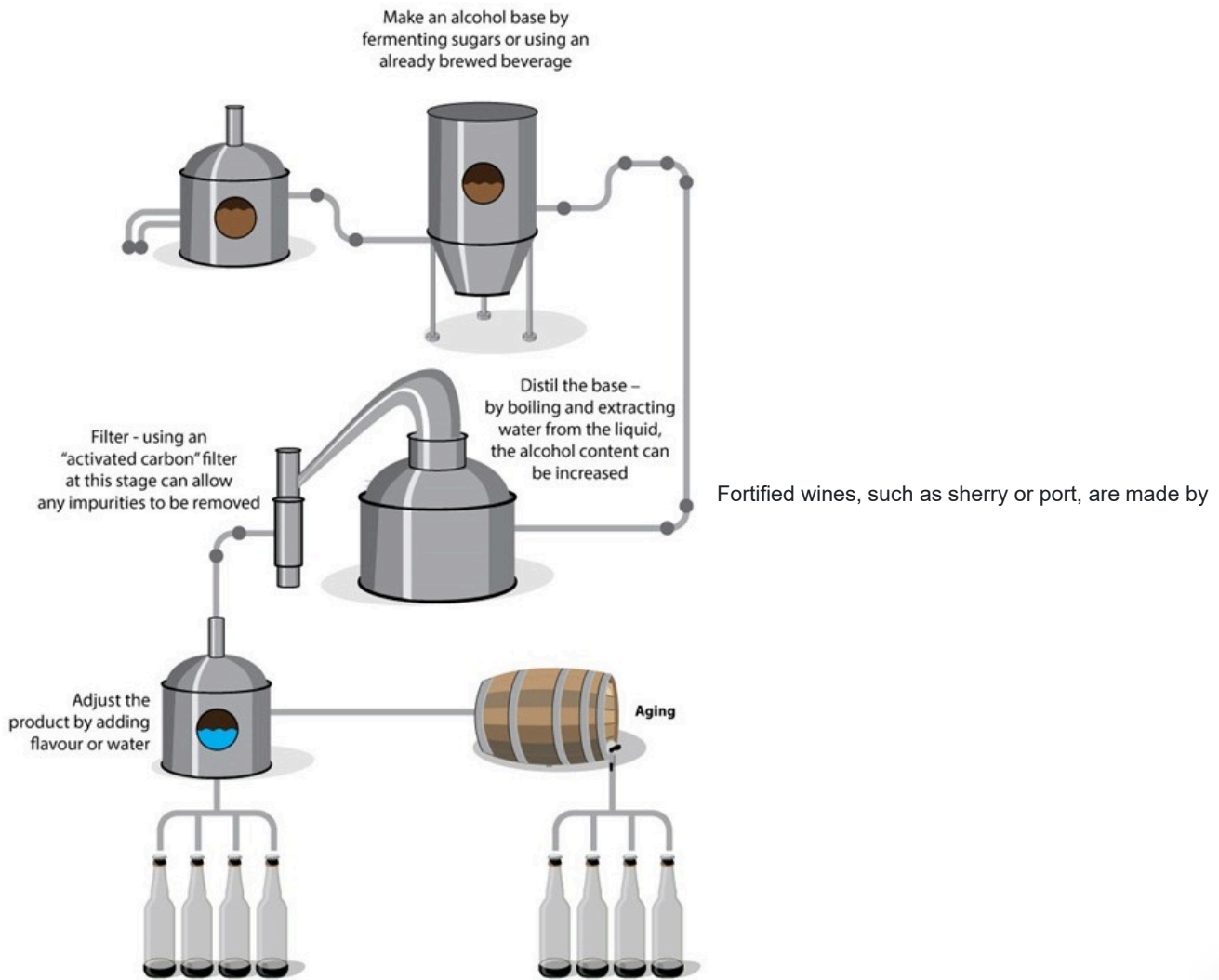
INTRODUCTION

All alcoholic drinks, spirits are made by fermentation. Spirits involve a further process of distillation where some of the water is removed to leave a stronger drink with a higher percentage of alcohol.

Distillation

Fermented alcoholic beverages such as beer, wine, or mead, tend to have lower alcohol content. When the beverage is distilled, the alcohol content is increased.





adding high alcohol content spirits to lower alcohol wine to create a medium alcohol level drink.

Some distilled fermented beverages may be sweetened to add taste. Sweetened spirits are called liqueurs and these strong drink are typically consumed after a meal. Many have flavours infused into them, so they do not necessarily taste sweet.

Distilled fermented beverages that are not sweetened, and which have alcohol content above 20% are called spirits. These include vodka, gin, brandy, whisky, and rum.

A Simplified Process

Make an alcohol base by fermenting sugars or using an already-brewed beverage

Distil the base – by boiling and extracting water from the liquid, the alcohol content can be increased

Filter - using an “activated carbon” filter at this stage can allow any impurities to be removed.

Adjust the product by adding flavour or water

SPIRITS

Traditionally spirits (eau-de-vie) are the product of distilled wine (brandy) but now they encompass a wide range of spirits distilled from fruit and cereals.

A spirit may be produced from any raw material which contains starch or sugar and has been fermented and distilled. The principle of distillation is that alcohol boils at a lower temperature than water, that is, at 78°C. The vapour is driven off and then passes through a condenser. After cooling, it returns to be collected as a spirit.

Classification of Spirits

Spirits are classified into six main groups:

- Brandy
- Whisky
- Gin
- Rum
- Vodka
- Tequila

All spirits and liqueurs are labelled with an alcohol by volume number (ABV). This system has replaced the older 'proof' value which used to be used as a measure of strength. Proof values were designated as a particular number of degrees proof. However, there was no universal agreement over what constituted the real proof value.

In the UK, proof was considered to be 1.75 times the ABV. In other words, a drink with an ABV of approximately 57.15% would be classed as 100 degrees proof. Proof was calculated in this way because spirits have a gravity which is 12/13 that of water.

In the United States, proof was set at 2 times ABV so that a drink with an ABV of 50% would be classed as 100° proof. As you can see, this was confusing. A person from the US might be forgiven for accidentally getting drunk consuming a spirit made in the UK using this old rating system.

Whilst all countries now use ABV to express the alcohol content of spirits, and other alcoholic drinks, in the UK and Europe the scale used is based on the International Organization of Legal Metrology (OIML) or the Gay-Lussac scale which expresses the amount in degrees rather than as a percentage.

Although no longer used commercially, some enthusiasts and home drinks makers might refer to or use proof as a measure of alcohol content. The UK system was called the Sikes Scale. To obtain a proof value you would have to measure it using a Sikes Hydrometer. This is a brass instrument with weights which when immersed in liquid gives a reading that, together with the temperature of a particular spirit and reference to a book of tables (Sikes Scale), enables the proof strength to be calculated. Using this scale, a bottle of whisky which is 70° proof is approximately 40% alcohol (since $100 \times 70/175 = 40\%$).

Spirit Groups

Brandy

Traditional brandy is the spirit that is obtained through distilling wine, although fruit brandies may be made from distilling fermented fruit juice. Brandy can be produced wherever wine is made, although the finest brandies are those of Cognac and Armagnac in France. These finer types are distilled twice before being aged in oak barrels. Distillation of Cognac is by the pot-still method which is carried out in two stages. Following distillation, it is stored in oak casks to allow the colour and familiar bouquet to develop. After blending, it is bottled and no further improvement takes place regardless of how long it is kept. The age and quality of Cognac is indicated by stars, where three stars refers to standard quality and five stars are awarded to a liqueur brandy. Cheaper brandies may be coloured with caramel to give the impression of aging, rather than being stored in oak.

Cognac is considered to be the best of all spirits by some, although many a whisky connoisseur would beg to differ. Its flavour is due to the grapes of a special vine, the '*Folle blanche*', which can also be used to make an average to rough-tasting white wine.

Whisky

Whisky is a drink made by distilling a 'wash' of malted barley or other cereals. Although whisky is distilled in many countries, 'Scotch' or 'Scottish Whisky' is better known and enjoys greater sales. Whisky that comes from Ireland or the United States is usually spelt "whiskey". In the United States, whisky is known as bourbon and is usually made from rye and maize (corn). Tennessee bourbon whiskies such as Jack Daniels are made using a slightly different process which applies a charcoal flavour. All bourbons have a slightly sweet taste. Canadian whisky is similar. Irish whisky tends to be drier with less sugar and a smoky or peaty aftertaste. Irish and Scottish whiskies are made from barley. The name 'malt whisky' refers to malted barley.

Pot stills are used in the making of Scottish and Irish whiskies whereby a single heated chamber is used, and the purified alcohol collects in a vessel. American whiskies are generally made using column stills which include several pot stills in series enclosed in a vertical tube. Whiskies are aged in wooden casks, typically oak, before bottling and do not age further once bottled.

There are a variety of whisky types including:

- Malt whisky (Scotland) - which contains at least 51% malted barley

- Rye whisky (mainly Canada) - made from at least 51% rye
- Bourbon whisky (United States) - made from a mash of barley and at least 51% maize
- Rye malt - made from at least 51% rye malt

Others include wheat whisky which contains at least 51% wheat and corn whisky which contains at least 80% corn. Blended whiskies in the United States can contain neutral grain spirits, colours, and flavours as well as straight whiskies. Blended whiskies in other countries only contain different whiskies and may even all be from the same distillery. A blended malt whisky only contains a mixture of single malt whiskies. Single malts come from the same distillery using the same malted grain but may be taken from several casks and mixed to achieve the desired taste. They are often given an age according to the age of the casks. Single-cask whiskies come from one cask and may be given a cask number. Cask-strength whiskies also come from one cask but are not usually diluted.

Gin

This spirit receives its name from the French word for its flavouring agent "Genevre" which means juniper. As well as juniper berries, other agents can be used to flavour gin and these include anise, angelica root and caraway seeds. Gin is regarded as a neutral spirit because it

is made from pure ethyl alcohol and then flavouring is added.

As such it can be made from any base product but usually it is made from rye, barley, or other grains. The spirit is cleaned and rectified by distilling or redistilling the grains. This process clears it of all of its by-products and yields a neutral or 'silent' spirit. This basic spirit can be flavoured, sweetened, or coloured. Gin requires no ageing. Whilst other spirits must be left to mature in casks or other containers, gin can be sold the day after it is made.

Rum

Rum is a spirit which is distilled from sugarcane. It can be made either from juices of sugarcane or molasses. Molasses is a rich, dark treacly substance which is drawn from sugarcane through processing it. Rum is often produced in countries where sugarcane is grown. Major rum producers are based in the Caribbean, South America, Pacific Islands and Australia.

The distillate is aged in oak barrels and various grades of rum may be produced. These include premium rums which are intended to be drunk neat, light rums which are used in cocktails, and dark or golden rums which are drunk neat or with mixers. Dark rums can vary in colour from red to brown or even black and are made from caramelised sugar or molasses. They are aged for longer than golden rums inside charred barrels. Golden rums have an amber colour and do not taste quite as strong as dark ones. Light rum has a light golden to clear appearance and is sweeter tasting. There are also flavoured rums which have been infused with flavours from fruits and spiced rums flavoured with pepper, cinnamon or other spices.

Vodka

Vodka is a traditional spirit that originates from Eastern Europe. The exact origin is unknown though it would appear to have its roots in Russia, Poland, and Czechoslovakia. It is made from wheat or rye grain. It can also be made from potatoes, corn, or fruits. Filtration is used extensively in making vodka. Whatever it is distilled from, the outcome is ethanol, a neutral spirit, which is tasteless and colourless. This spirit is mixed with water and sometimes flavourings or other impurities. Vodkas made from rye and wheat grains are usually considered to be the best quality.

In Britain it is made by means of a continuous still. It is further purified by filtration through activated charcoal. Although traditional vodka is left unflavoured, these days a variety of different flavourings may be used to give a subtle taste to the final product including ginger, chilli, vanilla, fruits, cinnamon and other spices.

Tequila

Tequila originates in Central America, specifically the city of Tequila in Mexico. It is made by fermenting the juice of the Blue Agave plant, *Agave tequilana*, which naturally grows in the surrounding area. Nowadays agaves may be grown and harvested in other regions, but each area will produce plants which impart a slightly different taste to the tequila. True tequila can only legally be made in several Mexican municipalities and it is widely regarded as a 'protected designation of origin' product in many countries.

LIQUEURS

Liqueurs are spirits that have been sweetened and flavoured with cream or a variety of herbs, flowers, fruits, seeds, or nuts. They were originally made in monasteries for medicinal purposes, and generally have a high content of alcohol. Whilst the base spirit from which they were made may have been aged for a long period of time, the liqueurs generally are only left for a resting period during production which allows the flavours to blend harmoniously.

Liqueurs usually have a sweet taste, and some are quite syrupy. They can be served on their own or mixed with cream or other mixers to form cocktails. They are also commonly used in cooking due to their pronounced sweetness. Typically, liqueurs are consumed as an aperitif before a meal or as a digestif after a meal when they are often served with coffee.

To make a liqueur, you first have to select a spirit base. The spirit which is used can be anything such as brandy, whisky, or rum, but more usually it is a neutral grain spirit so that it is easier to get the desired flavours into it. Once the base is chosen, you then have to decide how to extract the flavour from the ingredients you want to add.

There are at least four possible methods of production:

- **Maceration** - this is where the flavours from bruised herbs, flowers and fruits are infused into the base spirit over a long period of time, something akin to steeping a tea. It is a cold process and is useful if heat alters or damages the desired flavours.
- **Infusion** - this is the same as maceration except the liquid mixture is heated.
- **Distillation** - this is carried out by distilling a mixture of spirit and macerated flavouring agents. The centre part of the distillate forms a concentrate of flavour and it is this that is collected. Heat is used here so only flavours not affected by high temperatures are suited to this method.
- **Percolation** - this is an intensive method of extraction via distillation. It is also a hot process where heated pure spirit is passed through the botanicals in order to extract the flavour. The spirit liquid is allowed to drip through the flavouring material.

After the liqueur is prepared, it may be sweetened or coloured. The alcohol strength can also be amended before it is filtered and bottled. Sometimes a single liqueur containing all the components is made. In other cases, several different liqueurs are blended to create the desired end product.

Liqueur Groups

There are different ways to categorise liqueurs, and one such way is as follows:

Herb Liqueurs

These either contain one predominant flavour or many. For example, Kummel is made from caraway, and Chartreuse has as many as 130 herb flavours in it, whilst Galliano contains essences of 30 herbs.

Floral Liqueurs

These are made primarily from infused petals. An example is Rosilio made using rose petals.

Fruit Liqueurs

Such fruit as stone fruit, berries and citrus are used. Examples include: Manzana verde (apple), Limoncello (lemon), Peach Schnapps, Apricot Brandy and Cherry Brandy.

Berry Liqueurs

There are several well-known liqueurs with berry flavours such as Murtado (strawberry), Crème de Cassis (blackcurrant), and Chambord Black Raspberry (raspberry).

Cream Liqueurs

Bailey's Irish Cream is perhaps the most well-known in this group but there are many other examples, such as Advocaat (yellow coloured Dutch liqueur with eggs in it), Heather Cream (with malt whisky and cream), and Vana Tallin (with a rum base and vanilla and cinnamon flavours).

Chocolate Liqueurs

These liqueurs can be broken down further into Crème de Cacao, Chocolate, and Chocolate Cream. Some chocolate liqueurs include Mozart Black Chocolate, Mozart Gold Chocolate, Crème de Cacao, Liqueur Fogg and Godiva Liqueur.

Coffee Liqueurs

A number of coffee-flavoured liqueurs are popular, including Kahlua, Tia Maria, Sheridan's and Kamora.



APERITIFS

Aperitif is the term given to an alcoholic drink which is served before a meal to stimulate the appetite. It normally accompanies a small serving of food such as cheese, olives, and crackers. Vermouth and sherry are perhaps the two most common aperitifs however a dry type of champagne is also popular.

SHERRY

It is debatable whether sherry should be considered a spirit or a wine. In hotels and bars, it is often treated like a spirit by bartenders who open a bottle and leave it on the shelf with other spirits. It is, in fact, a fortified wine though and as any sherry connoisseur will tell you once opened it is best refrigerated and consumed within several days.

Most sherry is made from the juice of white Palomino Fino grapes which are grown in vineyards located in a small region of south west Spain called Cadiz. This region is loosely centred around the inland town of Jerez de la Frontera and extends to the coast. The name 'sherry' is derived from the town's name. It is fortified with a neutral spirit, or occasionally brandy, and blended using the solera system whereby a continuity of the particular style can be maintained almost indefinitely. This involves storing the same blend of mature wine in rows of barrels. A small amount of wine is taken from each barrel and used for bottling. The removed wine is then replaced by adding some young wine to the barrels and leaving this to mature. As such, sherry does not have a specific age since it is produced from a blend of aged wines.

There are three broad groups of sherry: Fino (pale and dry), dark (oxidised), and sweet (dry sherry and sweet wine blend).

Types of Sherry

The main types of sherry are:

Fino

This is the lightest type of sherry and has an ABV of 15 to 16%. It is very dry and pale and quite delicate with a clean taste. Finos are made beneath a layer of yeast called 'flor' so that they do not oxidise. They are often served as an aperitif and go well with salty snacks like peanuts. They can be chilled or served with ice.

Manzanilla

This is a type of fino which is made in the town of Sanlúcar de Barrameda. As such it is also light and dry with a pale colour. It can be served as an aperitif and is said to go well with seafood and tapas. Again, it can be served chilled or with ice.

Amontillado

This is made from fino but the flor which covers it has broken at some point so that it becomes oxidised. The result is a sherry with more body and a brownish colour which is less dry than usual finos and slightly nutty. It can be served as an aperitif making a good accompaniment to oily fish or chicken. It may be served chilled or with ice.

Oloroso

This is a richer, more full-bodied wine than Amontillado with a fine golden colour. Whereas with Amontillado the flor has broken naturally, in this case the flor is deliberately broken to encourage oxidation. It has a strong nutty flavour and a higher alcoholic content (18 to 19%) than the finos. It goes well with strong meats and mature cheeses.

Palo Cortado

This sherry begins like a fino with the flor intact but then becomes more like an amontillado as the flor breaks up. However, the final product is more similar to a richer oloroso. It is a full flavoured, dry wine similar in style to oloroso, but with a reddish-brown colour and less heaviness. It is best enjoyed by itself.

Pedro Ximénez

This is a sweet, dark, syrupy wine used as dessert sherry. It tastes a bit like toffee, chocolate, or molasses. It goes well with desserts or served on them, e.g., poured over vanilla ice cream.

Cream

These are often a sweet and smooth sherry somewhat like an oloroso but with sweet grapes blended in, however there are a variety of different types of cream sherries with some being more complex and fuller bodied.



Sweet (Jerez Dulce)

Sweet sherry is made from sun dried Pedro Ximénez or Muscatel grapes. This concentrates the sugars in the grape and produces sweet juice. These are dark and quite thick, syrupy wines.

VERMOUTH

Like liqueurs, some drinks are made by blending herbs with a wine base instead of spirits. Traditionally, these drinks were usually made for medicinal purposes. One of the most widely known modern versions is the aperitif wine called Vermouth. Vermouth is a white wine, or neutral grape wine, into which herbs are macerated to impart flavour. Additional alcohol is added. Variations include using a wine and spirit base or spirit only. After adding aromas and fortifying, sweeteners are also added. The result is a drink with a unique taste. The original Vermouth comes from the alpine regions of France and Italy and is said have been first made in Turin. Vermouth may be drunk as an aperitif or as a cocktail ingredient in such drinks as Martini, Rob Roy and the Manhattan.

Two types were formerly made by the French and the Italian producers:

- **Dry** - made in France and known as "French Sweet Vermouth"; a well-known French vermouth is Noilly Prat.
- **Sweet** - made in Italy and known simply as "Italian"; a well-known Italian vermouth is Cinzano.

In recent years other types have been brought to the market to satisfy demand, and these include red, amber, rosé, sweet white and extra dry white. Vermouth is most commonly used when mixed 'sweet' or 'dry' with gin to taste.

PORT

Like sherry, port is a fortified wine. However, port is heavier and sweeter than sherry. The original port is made from grapes which are grown in the Douro Valley in Portugal and fortified with brandy. From here, it is often shipped to the rest of the world through Oporto. However, port is also now produced in many countries around the world and some of these other countries produce excellent ports. Port is often consumed as an after-dinner drink or dessert wine. It goes well with a cheese board or nuts and chocolates. Tawny ports may be drunk as an aperitif. There are two main styles of port depending on how they are aged. That is, some are aged in the bottle through reductive ageing resulting in a slightly smoother wine. Others are aged in barrels through oxidative ageing making them slightly more syrupy. Both types lose colour over time.

Types of Port

There are several different types of port:

Vintage

Vintage ports are considered to be the best but only account for about 2 percent of port produced. It is the produce of one exceptional year which is declared within six months of harvesting the grapes. On average a vintage is only declared about three times in every ten-year period. The wine is firstly aged in the cask or barrels for up to a maximum of two and a half years. It is then further aged in bottles for ten to forty years. These ports retain their ruby colour and fresh taste even after many more years. A vintage port may have two dates: the year of the vintage and the year in which it was bottled.

Late bottled vintage (LBV) is bottled between its 4th and 6th year after the vintage. They were originally intended to be vintage ports but were set aside for storage during times of decreased demand. Some are filtered before bottling but others are not. Those that are filtered are ready for immediate drinking without decanting and come in resealable bottles. Some argue that this hinders flavour compared to non-filtered types which are sold in corked bottles and have to be decanted after opening.

Crusted

These ports are typically made from a blend of several different vintages, although they have been made from a single vintage sometimes. They are intended to be consumed earlier than vintage ports and are bottled after two years when they form a crust, although not as large a crust as vintage port. Crusted ports are aged after bottling for at least three years but often longer. They are sealed in corked bottles and are unfiltered, so they need decanting after opening.

Tawny

These are blended wines made from red grapes and aged in wooden barrels. Genuine tawny port is stored in the barrel for many years in which they oxidise and evaporate, processes which contribute to their tawny brown colour. Cheaper tawny ports may be made by mixing white port with ruby port. Aged tawny should have a pale, brownish red appearance and are fairly sweet and dry,



making them good dessert wines. They may be sold with an indication of age which is 10, 20, 30 or more than 40 years. This represents the minimum average age of the blends, not the actual age.

Ruby

This is the most standard type of red port. It is cheap and the most widely produced port. It is blended and similar to tawny, but it is stored in concrete or stainless-steel tanks for several years after fermentation. As such, it maintains its rich red colour and body because oxidation is inhibited. It is filtered before bottling and so does not get any better with age.

White

As the name suggests, white port is made from white grapes. It involves separating the stalks and skins from freshly pressed grapes. It can be sweet or dry. Aged white port is served as an aperitif or for enjoyment by itself and should be slightly chilled before drinking. Aged whites also gain darker colours with age and may resemble red ports. Other whites may be mixed in cocktails.

Rosé

These are not aged at all and are intended to be consumed fresh. They have a pink colour and smooth zesty taste. Rosés are a recent addition to ports.

Colheita

These are a type of tawny port made from a single vintage. However, they are not the same as vintage ports since they spend about 20 years in the barrel before being bottled (as opposed to just a few years).

Garrafeira

These are ports made from a single vintage which undergo oxidation in wooden barrels for several years (typically three to six) and then undergo reduction aging in large glass demijohns for a further number of years (at least eight). It is said by connoisseurs to have a slight bacon-like taste.

